Introduction to Mathematics Education Syllabus

MAT101 • Benjamin Patel

Education • Mathematics Teaching

1. Learning Objectives & Taxonomy

ID	Learning Objective	Bloom's Level
LO1	Analyze the historical and philosophical foundations of mathematics education to understand contemporary pedagogical debates.	Analyzing
LO2	Apply major learning theories such as Piaget and Vygotsky to design effective mathematics instruction.	Applying
LO3	Develop comprehensive mathematics curriculum units and instructional strategies considering curriculum models and standards.	Creating
LO4	Integrate digital tools and software effectively to enhance mathematical learning experiences.	Applying
LO5	Design appropriate formative and summative assessments along with rubrics to evaluate mathematical understanding.	Creating
LO6	Evaluate equity, diversity, and inclusion practices to ensure accessible mathematics learning environments for all learners.	Evaluating
LO7	Demonstrate advanced mathematical thinking through problem-solving, reasoning, and proof.	Evaluating

Overall Course Cognitive Depth: Analyzing, Evaluating, and Creating

2. Weekly Instructional Plan

Wk	Subject and Activities	Outcomes
1	**Introduction to Mathematics Education: Scope, Goals, and Current Landscape** Course Objectives / Learning Outcomes, Syllabus • Ice Breakers • Pre-Course Self-Assessment, Discussion Forums	LO1
2	**Historical and Philosophical Foundations of Mathematics Education** Video Lectures, Reading List • Group Discussions • Article Analysis	LO1
3	**Major Learning Theories Applied to Mathematics (e.g., Piaget, Vygotsky, Constructivism)** Role Playing, Face to Face Lectures • Case Study • Quizzes	LO2
4	**Curriculum Development in Mathematics Education: Principles and Practices** Collaborative Teaching Assignment • Group Exercises	LO3
5	**Technology Integration in Mathematics Education: Digital Tools, Software, and Online Resources** Discussion Forums, Video Lectures • Tool Discovery Assignment, Workshops • Assignments / Homework, Quizzes	LO4
6	**Assessment for Learning in Mathematics: Formative and Summative Approaches** Practice Exams • Peer Feedback • Rubrics, Assignments / Homework	LO5
7	**Instructional Strategies and Pedagogical Approaches for Teaching Mathematics** Peer Instruction, Face to Face Lectures • Think-Pair-Share, Group Discussions • One Minute Paper, Assignments / Homework	LO3, LO2
8	**Addressing Equity, Diversity, and Inclusion in Mathematics Education** Invited Speakers, Panel Discussions • Socratic Seminar • Midterm Exam	LO6
9	**Developing Mathematical Thinking: Problem-Solving, Reasoning, and Proof** Collaborative Study Sessions, Mathematical Proofs • Challenge Problems	LO7
10	**Teaching Fractions, Ratios, Proportions, and Early Algebra in School Mathematics** Recitation Sessions, Lecture Notes • Concept Maps • Challenge Problems, Problem Sets, Quizzes	LO7, LO3
11	**Affective Aspects of Mathematics Education: Beliefs, Attitudes, and Motivation** Storytelling • Reflection Essays	LO2
12	**Classroom Management and Creating a Positive Learning Environment for Mathematics** Case Study • Role-Playing	LO6

Wk	Subject and Activities	Outcomes
13	**Connecting Mathematics to Real-World Contexts and Interdisciplinary Applications** Experiential Learning, Real-World Cases • Sustainable Development Goals Project	LO3, LO7
14	**Introduction to Research Methods in Mathematics Education** Research Report • Seminars • Individual Projects, Literature Reviews	LO1, LO3
15	**Contemporary Issues and Future Directions in Mathematics Education** Webinars • Open Discussions • Critiques	LO1, LO6
16	**Final Exam** Practice Exams • Study Circles • Final Exam	—

3. Skill Acquisitions by Category

Domain	Specific Skill	Related Wk
Hard	Curriculum Design	4
	Digital Tool Integration	7
	Assessment Design	6
	Mathematical Proof Construction	9
Soft	Classroom Management	12
	Inclusive Teaching and Equity Advocacy	8
	Peer Feedback and Collaboration	4, 6
Meta	Critical Reflection and Research Inquiry	2, 14, 15
	Problem-Solving and Heuristic Reasoning	9, 13

4. Proposed Micro-Credential Framework

Credential Name	Dimensions	Depth	Align.	Mapping
Certified Mathematics Curriculum Innovator	Application	Proficiency	W4, 5, 7	#CurriculumDesign #DigitalToolIntegration #PedagogicalDevelopment
Inclusive and Equitable Math Educator	Human Dimension	Mastery	W8, 11, 12	#InclusiveTeaching #ClassroomManagement #StudentMotivation
Advanced Math Assessment Specialist	Integration	Proficiency	W6, 16	#AssessmentDesign #EducationalMeasurement #AcademicAssessment

5. Assessment & Mapping

Activity Name	Week	Weight	Bloom's	Outcomes
Pre-Course Self-Assessment	1	5%	Remembering	LO1
Quizzes	3	10%	Applying	LO2
Collab. Teaching	4	15%	Creating	LO3
Midterm Exam	8	25%	Evaluating	LO1, LO2, LO6
Assignments	6	10%	Creating	LO5
Problem Sets	10	10%	Analyzing	LO7
Final Exam	16	25%	Evaluating	LO1, LO2, LO3, LO4, LO5, LO6, LO7